PG No. 39 $S_4(c)$ $\bar{4}$ [tetragonal] (internal axial octupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(3,3)}[g](B)$

** symmetry

1

** expression

$$\frac{\sqrt{105}G_{xyz}xyz}{7}+\frac{\sqrt{7}G_x^\alpha x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{105}G_x^\beta x\left(y-z\right)\left(y+z\right)}{14}-\frac{\sqrt{7}G_y^\alpha y\left(3x^2-2y^2+3z^2\right)}{14}$$
$$-\frac{\sqrt{105}G_y^\beta y\left(x-z\right)\left(x+z\right)}{14}-\frac{\sqrt{7}G_z^\alpha z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}G_z^\beta z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(3,1)}[g](A)$

** symmetry

z

** expression

$$\frac{\sqrt{21}G_{xyz}xy}{7}-\frac{3\sqrt{35}G_x^\alpha xz}{35}-\frac{\sqrt{21}G_x^\beta xz}{7}-\frac{3\sqrt{35}G_y^\alpha yz}{35}+\frac{\sqrt{21}G_y^\beta yz}{7}-\frac{3\sqrt{35}G_z^\alpha\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}G_z^\beta\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](A)$

** symmetry

z

** expression

$$-\frac{\sqrt{35}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{5\sqrt{21}G_x^\alpha xz\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta xz\left(2x^2+9y^2-5z^2\right)}{28}-\frac{5\sqrt{21}G_y^\alpha yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$-\frac{\sqrt{35}G_y^\beta yz\left(9x^2+2y^2-5z^2\right)}{28}+\frac{\sqrt{21}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}G_z^\beta\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](B)$

** symmetry

z

** expression

$$-\frac{\sqrt{105}G_{xyz}z\left(x-y\right)\left(x+y\right)}{14}+\frac{3\sqrt{7}G_x^\alpha y\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}G_x^\beta y\left(2x^2-y^2+z^2\right)}{28}$$
$$+\frac{3\sqrt{7}G_y^\alpha x\left(x^2-4y^2+z^2\right)}{28}+\frac{\sqrt{105}G_y^\beta x\left(x^2-2y^2-z^2\right)}{28}+\frac{\sqrt{105}G_z^\beta xyz}{7}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{210}iG_{xyz}\left(x^2y+ixy^2-ixz^2-yz^2\right)}{28}-\frac{3\sqrt{14}iG_x^\alpha z\left(4x^2-y^2-z^2\right)}{56}-\frac{\sqrt{210}iG_x^\beta z\left(2x^2+4ixy+y^2-z^2\right)}{56}-\frac{3\sqrt{14}G_y^\alpha z\left(x^2-4y^2+z^2\right)}{56}$$
$$-\frac{\sqrt{210}G_y^\beta z\left(x^2-4ixy+2y^2-z^2\right)}{56}-\frac{3\sqrt{14}iG_z^\alpha\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{56}+\frac{\sqrt{210}iG_z^\beta\left(x^3+ix^2y-xy^2-2xz^2-iy^3+2iyz^2\right)}{56}$$

$\vec{\mathbb{G}}_1^{(3,1)}[g](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{42}iG_{xyz}z\left(x+iy\right)}{14}+\frac{3\sqrt{70}G_x^\alpha\left(2x^2+2ixy-y^2-z^2\right)}{140}-\frac{\sqrt{42}iG_x^\beta\left(2xy+iy^2-iz^2\right)}{28}$$
$$+\frac{3\sqrt{70}iG_y^\alpha\left(x^2+2ixy-2y^2+z^2\right)}{140}+\frac{\sqrt{42}iG_y^\beta\left(x^2+2ixy-z^2\right)}{28}-\frac{3\sqrt{70}G_z^\alpha z\left(x-iy\right)}{70}+\frac{\sqrt{42}G_z^\beta z\left(x+iy\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{70}iG_{xyz}z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{28}+\frac{\sqrt{42}G_x^{\alpha}\left(8x^4-20ix^3y-24x^2y^2-24x^2z^2+15ixy^3+15ixyz^2+3y^4+6y^2z^2+3z^4\right)}{168}$$
$$+\frac{\sqrt{70}iG_x^{\beta}\left(2x^3y-6ix^2y^2+6ix^2z^2-5xy^3+9xyz^2+iy^4-iz^4\right)}{56}$$
$$-\frac{\sqrt{42}iG_y^{\alpha}\left(3x^4-15ix^3y-24x^2y^2+6x^2z^2+20ixy^3-15ixyz^2+8y^4-24y^2z^2+3z^4\right)}{168}$$
$$-\frac{\sqrt{70}iG_y^{\beta}\left(x^4-5ix^3y-6x^2y^2+2ixy^3+9ixyz^2+6y^2z^2-z^4\right)}{56}$$
$$-\frac{5\sqrt{42}G_z^{\alpha}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}G_z^{\beta}z\left(5x^3-9ix^2y-9xy^2-2xz^2+5iy^3-2iyz^2\right)}{56}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{210}iG_{xyz}\left(x^2y-ixy^2+ixz^2-yz^2\right)}{28}+\frac{3\sqrt{14}iG_x^{\alpha}z\left(4x^2-y^2-z^2\right)}{56}+\frac{\sqrt{210}iG_x^{\beta}z\left(2x^2-4ixy+y^2-z^2\right)}{56}-\frac{3\sqrt{14}G_y^{\alpha}z\left(x^2-4y^2+z^2\right)}{56}$$
$$-\frac{\sqrt{210}G_y^{\beta}z\left(x^2+4ixy+2y^2-z^2\right)}{56}+\frac{3\sqrt{14}iG_z^{\alpha}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{210}iG_z^{\beta}\left(x^3-ix^2y-xy^2-2xz^2+iy^3-2iyz^2\right)}{56}$$

$\vec{\mathbb{G}}_1^{(3,1)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{42}iG_{xyz}z\left(x-iy\right)}{14}+\frac{3\sqrt{70}G_x^{\alpha}\left(2x^2-2ixy-y^2-z^2\right)}{140}+\frac{\sqrt{42}iG_x^{\beta}\left(2xy-iy^2+iz^2\right)}{28}$$
$$-\frac{3\sqrt{70}iG_y^{\alpha}\left(x^2-2ixy-2y^2+z^2\right)}{140}-\frac{\sqrt{42}iG_y^{\beta}\left(x^2-2ixy-z^2\right)}{28}-\frac{3\sqrt{70}G_z^{\alpha}z\left(x+iy\right)}{70}+\frac{\sqrt{42}G_z^{\beta}z\left(x-iy\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{70}iG_{xyz}z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{28}+\frac{\sqrt{42}G_x^{\alpha}\left(8x^4+20ix^3y-24x^2y^2-24x^2z^2-15ixy^3-15ixyz^2+3y^4+6y^2z^2+3z^4\right)}{168}$$
$$-\frac{\sqrt{70}iG_x^{\beta}\left(2x^3y+6ix^2y^2-6ix^2z^2-5xy^3+9xyz^2-iy^4+iz^4\right)}{56}$$
$$+\frac{\sqrt{42}iG_y^{\alpha}\left(3x^4+15ix^3y-24x^2y^2+6x^2z^2-20ixy^3+15ixyz^2+8y^4-24y^2z^2+3z^4\right)}{168}$$
$$+\frac{\sqrt{70}iG_y^{\beta}\left(x^4+5ix^3y-6x^2y^2-2ixy^3-9ixyz^2+6y^2z^2-z^4\right)}{56}$$
$$-\frac{5\sqrt{42}G_z^{\alpha}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}G_z^{\beta}z\left(5x^3+9ix^2y-9xy^2-2xz^2-5iy^3+2iyz^2\right)}{56}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(3,0)}[g](A)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{210}G_{xyz}\left(x-y\right)\left(x+y\right)}{28}-\frac{3\sqrt{14}G_x^{\alpha}yz}{28}-\frac{\sqrt{210}G_x^{\beta}yz}{28}+\frac{3\sqrt{14}G_y^{\alpha}xz}{28}-\frac{\sqrt{210}G_y^{\beta}xz}{28}+\frac{\sqrt{210}G_z^{\beta}xy}{14}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](A)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{105}G_{xyz}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{42}+\frac{5\sqrt{7}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}G_x^{\beta}yz\left(12x^2-9y^2+5z^2\right)}{84}$$
$$+\frac{5\sqrt{7}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{28}+\frac{\sqrt{105}G_y^{\beta}xz\left(9x^2-12y^2-5z^2\right)}{84}-\frac{\sqrt{105}G_z^{\beta}xy\left(x^2+y^2-6z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{7}G_x^{\alpha}x}{14}-\frac{\sqrt{105}G_x^{\beta}x}{42}-\frac{3\sqrt{7}G_y^{\alpha}y}{14}-\frac{\sqrt{105}G_y^{\beta}y}{42}+\frac{\sqrt{105}G_z^{\beta}z}{21}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z}{21}-\frac{\sqrt{7}G_x^{\alpha}y}{7}+\frac{\sqrt{105}G_x^{\beta}y}{21}-\frac{\sqrt{7}G_y^{\alpha}x}{7}-\frac{\sqrt{105}G_y^{\beta}x}{21}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}G_x^{\alpha}x\left(4x^2-y^2-11z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{7}G_y^{\alpha}y\left(x^2-4y^2+11z^2\right)}{28}$$
$$-\frac{\sqrt{105}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{84}-\frac{5\sqrt{7}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{42}+\frac{\sqrt{7}G_x^{\alpha}y\left(4x^2-y^2-z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}y\left(6x^2+y^2-9z^2\right)}{84}$$
$$-\frac{\sqrt{7}G_y^{\alpha}x\left(x^2-4y^2+z^2\right)}{28}-\frac{\sqrt{105}G_y^{\beta}x\left(x^2+6y^2-9z^2\right)}{84}-\frac{5\sqrt{7}G_z^{\alpha}xyz}{7}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{2310}G_{xyz}xyz\left(x-y\right)\left(x+y\right)}{44}+\frac{5\sqrt{154}G_x^{\alpha}x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_x^{\beta}x\left(2x^4-73x^2y^2+53x^2z^2+51y^4-87y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_y^{\alpha}y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_y^{\beta}y\left(51x^4-73x^2y^2-87x^2z^2+2y^4+53y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$
$$+\frac{\sqrt{2310}G_z^{\beta}z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2310}G_{xyz}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{462}+\frac{5\sqrt{154}G_x^{\alpha}y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{2310}G_x^{\beta}y\left(12x^4-46x^2y^2+66x^2z^2+5y^4-4y^2z^2-9z^4\right)}{924}+\frac{5\sqrt{154}G_y^{\alpha}x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{2310}G_y^{\beta}x\left(5x^4-46x^2y^2-4x^2z^2+12y^4+66y^2z^2-9z^4\right)}{924}-\frac{5\sqrt{154}G_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{3\sqrt{2310}G_z^{\beta}xyz\left(x-y\right)\left(x+y\right)}{44}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](B,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{70}G_{xyz}\left(x^2+y^2-2z^2\right)}{28}-\frac{\sqrt{42}G_x^{\alpha}yz}{28}+\frac{3\sqrt{70}G_x^{\beta}yz}{28}-\frac{\sqrt{42}G_y^{\alpha}xz}{28}-\frac{3\sqrt{70}G_y^{\beta}xz}{28}+\frac{\sqrt{42}G_z^{\alpha}xy}{14}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](B,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_x^{\alpha}xz}{7}+\frac{\sqrt{42}G_y^{\alpha}yz}{7}-\frac{\sqrt{42}G_z^{\alpha}\left(x-y\right)\left(x+y\right)}{28}+\frac{\sqrt{70}G_z^{\beta}\left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](B,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{35}G_{xyz}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{42}+\frac{5\sqrt{21}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{84}+\frac{\sqrt{35}G_x^{\beta}yz\left(36x^2+y^2-13z^2\right)}{84}$$
$$-\frac{5\sqrt{21}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{84}-\frac{\sqrt{35}G_y^{\beta}xz\left(x^2+36y^2-13z^2\right)}{84}+\frac{5\sqrt{21}G_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{42}-\frac{\sqrt{35}G_z^{\beta}xy\left(x-y\right)\left(x+y\right)}{6}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](B,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{35}G_{xyz}xy\left(x-y\right)\left(x+y\right)}{6}-\frac{5\sqrt{21}G_x^{\alpha}xz\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}G_x^{\beta}xz\left(2x^2-3y^2-z^2\right)}{12}-\frac{5\sqrt{21}G_y^{\alpha}yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^{\beta}yz\left(3x^2-2y^2+z^2\right)}{12}-\frac{5\sqrt{21}G_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{84}+\frac{\sqrt{35}G_z^{\beta}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{84}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](B)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^{\alpha}x}{14}-\frac{\sqrt{35}G_x^{\beta}x}{14}-\frac{\sqrt{21}G_y^{\alpha}y}{14}+\frac{\sqrt{35}G_y^{\beta}y}{14}+\frac{\sqrt{21}G_z^{\alpha}z}{7}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](B)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^{\alpha}x\left(4x^2-21y^2+9z^2\right)}{84}+\frac{\sqrt{35}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{28}+\frac{\sqrt{21}G_y^{\alpha}y\left(21x^2-4y^2-9z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{28}-\frac{\sqrt{21}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](B)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{3\sqrt{770}G_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{462}G_x^{\alpha}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{1848}$$
$$-\frac{\sqrt{770}G_x^{\beta}x\left(2x^4+11x^2y^2-31x^2z^2+9y^4-87y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_y^{\alpha}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{1848}+\frac{\sqrt{770}G_y^{\beta}y\left(9x^4+11x^2y^2-87x^2z^2+2y^4-31y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}G_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E^{(a)})$

** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{21}iG_x^{\alpha}\left(4xy+iy^2-iz^2\right)}{28}-\frac{\sqrt{35}G_x^{\beta}\left(2x^2-y^2-z^2\right)}{28}+\frac{\sqrt{21}iG_y^{\alpha}\left(x^2+4ixy-z^2\right)}{28}+\frac{\sqrt{35}iG_y^{\beta}\left(x^2-2y^2+z^2\right)}{28}+\frac{\sqrt{21}G_z^{\alpha}z\left(x-iy\right)}{7}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E^{(a)})$

** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{70}iG_{xyz}z\left(x^3-xz^2+iy^3-iyz^2\right)}{12}+\frac{5\sqrt{42}iG_x^{\alpha}\left(4x^3y-6ix^2y^2+6ix^2z^2-3xy^3-3xyz^2+iy^4-iz^4\right)}{168}$$
$$+\frac{\sqrt{70}G_x^{\beta}\left(4x^4-14ix^3y-12x^2y^2-12x^2z^2+7ixy^3+21ixyz^2+5y^4-18y^2z^2+5z^4\right)}{168}$$
$$+\frac{5\sqrt{42}iG_y^{\alpha}\left(x^4-3ix^3y-6x^2y^2+4ixy^3-3ixyz^2+6y^2z^2-z^4\right)}{168}$$
$$+\frac{\sqrt{70}iG_y^{\beta}\left(5x^4-7ix^3y-12x^2y^2-18x^2z^2+14ixy^3-21ixyz^2+4y^4-12y^2z^2+5z^4\right)}{168}$$
$$-\frac{5\sqrt{42}G_z^{\alpha}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}G_z^{\beta}z\left(x^3+3ix^2y+3xy^2-2xz^2+iy^3-2iyz^2\right)}{24}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E^{(a)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{210}G_{xyz}\left(x-iy\right)}{42}+\frac{\sqrt{14}iG_x^{\alpha}z}{14}+\frac{\sqrt{210}iG_x^{\beta}z}{42}-\frac{\sqrt{14}G_y^{\alpha}z}{14}+\frac{\sqrt{210}G_y^{\beta}z}{42}+\frac{\sqrt{14}iG_z^{\alpha}\left(x+iy\right)}{14}-\frac{\sqrt{210}iG_z^{\beta}\left(x-iy\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E^{(a)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{210}G_{xyz}\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{84}-\frac{\sqrt{14}iG_x^{\alpha}z\left(4x^2-20ixy-y^2-z^2\right)}{56}$$
$$+\frac{\sqrt{210}iG_x^{\beta}z\left(6x^2-9y^2+z^2\right)}{168}-\frac{\sqrt{14}G_y^{\alpha}z\left(x^2-20ixy-4y^2+z^2\right)}{56}-\frac{\sqrt{210}G_y^{\beta}z\left(9x^2-6y^2-z^2\right)}{168}$$
$$+\frac{\sqrt{14}iG_z^{\alpha}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{210}iG_z^{\beta}\left(x^3+9ix^2y-9xy^2+6xz^2-iy^3-6iyz^2\right)}{168}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E^{(a)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{1155}G_{xyz}\left(x^5+6ix^4y-5x^3y^2-5x^3z^2+5ix^2y^3-51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4-iy^5+5iy^3z^2+6iyz^4\right)}{462}$$
$$-\frac{5\sqrt{77}iG_x^\alpha z\left(8x^4+14ix^3y-12x^2y^2-12x^2z^2-7ixy^3-7ixyz^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{1155}iG_x^\beta z\left(12x^4+66x^2y^2-46x^2z^2+63ixy^3-63ixyz^2-9y^4-4y^2z^2+5z^4\right)}{924}$$
$$+\frac{5\sqrt{77}G_y^\alpha z\left(x^4+7ix^3y-12x^2y^2+2x^2z^2-14ixy^3+7ixyz^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{1155}G_y^\beta z\left(9x^4+63ix^3y-66x^2y^2+4x^2z^2-63ixyz^2-12y^4+46y^2z^2-5z^4\right)}{924}$$
$$-\frac{5\sqrt{77}iG_z^\alpha\left(x+iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}$$
$$+\frac{\sqrt{1155}iG_z^\beta\left(5x^5+9ix^4y-4x^3y^2-46x^3z^2+4ix^2y^3-66ix^2yz^2-9xy^4+66xy^2z^2+12xz^4-5iy^5+46iy^3z^2-12iyz^4\right)}{924}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E^{(b)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{21}iG_x^\alpha\left(4xy-iy^2+iz^2\right)}{28}-\frac{\sqrt{35}G_x^\beta\left(2x^2-y^2-z^2\right)}{28}-\frac{\sqrt{21}iG_y^\alpha\left(x^2-4ixy-z^2\right)}{28}-\frac{\sqrt{35}iG_y^\beta\left(x^2-2y^2+z^2\right)}{28}+\frac{\sqrt{21}G_z^\alpha z\left(x+iy\right)}{7}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E^{(b)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{70}iG_{xyz}z\left(x^3-xz^2-iy^3+iyz^2\right)}{12}-\frac{5\sqrt{42}iG_x^\alpha\left(4x^3y+6ix^2y^2-6ix^2z^2-3xy^3-3xyz^2-iy^4+iz^4\right)}{168}$$
$$+\frac{\sqrt{70}G_x^\beta\left(4x^4+14ix^3y-12x^2y^2-12x^2z^2-7ixy^3-21ixyz^2+5y^4-18y^2z^2+5z^4\right)}{168}$$
$$-\frac{5\sqrt{42}iG_y^\alpha\left(x^4+3ix^3y-6x^2y^2-4ixy^3+3ixyz^2+6y^2z^2-z^4\right)}{168}$$
$$-\frac{\sqrt{70}iG_y^\beta\left(5x^4+7ix^3y-12x^2y^2-18x^2z^2-14ixy^3+21ixyz^2+4y^4-12y^2z^2+5z^4\right)}{168}$$
$$-\frac{5\sqrt{42}G_z^\alpha z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}G_z^\beta z\left(x^3-3ix^2y+3xy^2-2xz^2-iy^3+2iyz^2\right)}{24}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{210}G_{xyz}\left(x+iy\right)}{42}-\frac{\sqrt{14}iG_x^\alpha z}{14}-\frac{\sqrt{210}iG_x^\beta z}{42}-\frac{\sqrt{14}G_y^\alpha z}{14}+\frac{\sqrt{210}G_y^\beta z}{42}-\frac{\sqrt{14}iG_z^\alpha\left(x-iy\right)}{14}+\frac{\sqrt{210}iG_z^\beta\left(x+iy\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{210}G_{xyz}\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{84}+\frac{\sqrt{14}iG_x^\alpha z\left(4x^2+20ixy-y^2-z^2\right)}{56}$$
$$-\frac{\sqrt{210}iG_x^\beta z\left(6x^2-9y^2+z^2\right)}{168}-\frac{\sqrt{14}G_y^\alpha z\left(x^2+20ixy-4y^2+z^2\right)}{56}-\frac{\sqrt{210}G_y^\beta z\left(9x^2-6y^2-z^2\right)}{168}$$
$$-\frac{\sqrt{14}iG_z^\alpha\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{56}+\frac{\sqrt{210}iG_z^\beta\left(x^3-9ix^2y-9xy^2+6xz^2+iy^3+6iyz^2\right)}{168}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E^{(b)})$

** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{1155}G_{xyz}\left(x^5-6ix^4y-5x^3y^2-5x^3z^2-5ix^2y^3+51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4+iy^5-5iy^3z^2-6iyz^4\right)}{462}$$
$$+\frac{5\sqrt{77}iG_x^\alpha z\left(8x^4-14ix^3y-12x^2y^2-12x^2z^2+7ixy^3+7ixyz^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{1155}iG_x^\beta z\left(12x^4+66x^2y^2-46x^2z^2-63ixy^3+63ixyz^2-9y^4-4y^2z^2+5z^4\right)}{924}$$
$$+\frac{5\sqrt{77}G_y^\alpha z\left(x^4-7ix^3y-12x^2y^2+2x^2z^2+14ixy^3-7ixyz^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{1155}G_y^\beta z\left(9x^4-63ix^3y-66x^2y^2+4x^2z^2+63ixyz^2-12y^4+46y^2z^2-5z^4\right)}{924}$$
$$+\frac{5\sqrt{77}iG_z^\alpha\left(x-iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}$$
$$-\frac{\sqrt{1155}iG_z^\beta\left(5x^5-9ix^4y-4x^3y^2-46x^3z^2-4ix^2y^3+66ix^2yz^2-9xy^4+66xy^2z^2+12xz^4+5iy^5-46iy^3z^2+12iyz^4\right)}{924}$$

* Harmonics for rank 3

$\bar{\mathbb{Q}}_3^{(3,-2)}[g](A,1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_x^\beta x}{3}-\frac{\sqrt{3}G_y^\beta y}{3}-\frac{\sqrt{3}G_z^\beta z}{3}$$

$\bar{\mathbb{Q}}_3^{(3,-2)}[g](A,2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_{xyz}z}{3}+\frac{\sqrt{5}G_x^\alpha y}{4}+\frac{\sqrt{3}G_x^\beta y}{12}+\frac{\sqrt{5}G_y^\alpha x}{4}-\frac{\sqrt{3}G_y^\beta x}{12}$$

$\bar{\mathbb{Q}}_3^{(3,0)}[g](A,1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{10}G_x^\alpha x\left(y-z\right)\left(y+z\right)}{4}-\frac{\sqrt{6}G_x^\beta x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_y^\alpha y\left(x-z\right)\left(x+z\right)}{4}$$
$$+\frac{\sqrt{6}G_y^\beta y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_z^\alpha z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}G_z^\beta z\left(3x^2+3y^2-2z^2\right)}{12}$$

$\bar{\mathbb{Q}}_3^{(3,0)}[g](A,2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_x^\alpha y\left(x-z\right)\left(x+z\right)}{4}+\frac{\sqrt{6}G_x^\beta y\left(3x^2-2y^2+3z^2\right)}{12}$$
$$+\frac{\sqrt{10}G_y^\alpha x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{6}G_y^\beta x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_z^\alpha xyz}{2}$$

$\bar{\mathbb{Q}}_3^{(3,2)}[g](A,1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{7\sqrt{55}G_x^\alpha x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44}+\frac{\sqrt{33}G_x^\beta x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}$$
$$+\frac{7\sqrt{55}G_y^\alpha y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44}+\frac{\sqrt{33}G_y^\beta y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}$$
$$-\frac{7\sqrt{55}G_z^\alpha z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{33}G_z^\beta z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](A,2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{33}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132} + \frac{\sqrt{55}G_x^\alpha y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$
$$-\frac{\sqrt{33}G_x^\beta y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264} + \frac{\sqrt{55}G_y^\alpha x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$
$$+\frac{\sqrt{33}G_y^\beta x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264} + \frac{7\sqrt{55}G_z^\alpha xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](A)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$G_z^\alpha$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](A)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-G_{xyz}xy - \frac{\sqrt{15}G_x^\alpha xz}{20} - \frac{G_x^\beta xz}{4} - \frac{\sqrt{15}G_y^\alpha yz}{20} + \frac{G_y^\beta yz}{4} - \frac{\sqrt{15}G_z^\alpha\left(x^2+y^2-2z^2\right)}{15} - \frac{G_z^\beta(x-y)(x+y)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](A)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{330}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{132} - \frac{5\sqrt{22}G_x^\alpha xz\left(x^2-6y^2+z^2\right)}{44} + \frac{\sqrt{330}G_x^\beta xz\left(9x^2-12y^2-5z^2\right)}{132} + \frac{5\sqrt{22}G_y^\alpha yz\left(6x^2-y^2-z^2\right)}{44}$$
$$+\frac{\sqrt{330}G_y^\beta yz\left(12x^2-9y^2+5z^2\right)}{132} + \frac{\sqrt{22}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{88} + \frac{\sqrt{330}G_z^\beta(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](A)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{7\sqrt{715}G_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572} - \frac{35\sqrt{429}G_x^\alpha xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_x^\beta xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144}$$
$$+\frac{35\sqrt{429}G_y^\alpha yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144} + \frac{7\sqrt{715}G_y^\beta yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}$$
$$-\frac{5\sqrt{429}G_z^\alpha\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_z^\beta(x-y)(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{3}G_x^\alpha y}{4} - \frac{\sqrt{5}G_x^\beta y}{4} + \frac{\sqrt{3}G_y^\alpha x}{4} - \frac{\sqrt{5}G_y^\beta x}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{10}G_{xyz}z\,(x-y)\,(x+y)}{4}-\frac{\sqrt{6}G_x^{\alpha}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_x^{\beta}y\,(x-z)\,(x+z)}{4}$$
$$-\frac{\sqrt{6}G_y^{\alpha}x\left(2x^2-3y^2-3z^2\right)}{12}+\frac{\sqrt{10}G_y^{\beta}x\,(y-z)\,(y+z)}{4}+\frac{\sqrt{10}G_z^{\beta}xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{55}G_{xyz}z\,(x-y)\,(x+y)\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{33}G_x^{\alpha}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}$$
$$+\frac{\sqrt{55}G_x^{\beta}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}-\frac{5\sqrt{33}G_y^{\alpha}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}$$
$$-\frac{\sqrt{55}G_y^{\beta}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}-\frac{7\sqrt{55}G_z^{\beta}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_{xyz}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$G_z^{\beta}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-G_x^{\alpha}yz-G_y^{\alpha}xz-G_z^{\alpha}xy$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_x^{\alpha}xz}{4}-\frac{\sqrt{15}G_x^{\beta}xz}{4}-\frac{G_y^{\alpha}yz}{4}-\frac{\sqrt{15}G_y^{\beta}yz}{4}-\frac{G_z^{\alpha}\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{7\sqrt{22}G_{xyz}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{66}-\frac{\sqrt{330}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{35\sqrt{22}G_x^{\beta}yz\,(y-z)\,(y+z)}{132}$$
$$+\frac{\sqrt{330}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{132}-\frac{35\sqrt{22}G_y^{\beta}xz\,(x-z)\,(x+z)}{132}+\frac{\sqrt{330}G_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{132}+\frac{35\sqrt{22}G_z^{\beta}xy\,(x-y)\,(x+y)}{132}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B,2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{35\sqrt{22}G_{xyz}xy\left(x-y\right)\left(x+y\right)}{132} + \frac{\sqrt{330}G_x^\alpha xz\left(5x^2+12y^2-9z^2\right)}{132} - \frac{5\sqrt{22}G_x^\beta xz\left(x^2-6y^2+z^2\right)}{132} - \frac{\sqrt{330}G_y^\alpha yz\left(12x^2+5y^2-9z^2\right)}{132}$$
$$+ \frac{5\sqrt{22}G_y^\beta yz\left(6x^2-y^2-z^2\right)}{132} + \frac{\sqrt{330}G_z^\alpha\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{264} + \frac{7\sqrt{22}G_z^\beta\left(x^4-18x^2y^2+12x^2z^2+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B,1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{429}G_{xyz}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{286}$$
$$+ \frac{7\sqrt{715}G_x^\alpha yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572} + \frac{21\sqrt{429}G_x^\beta yz\left(y-z\right)\left(y+z\right)\left(10x^2-y^2-z^2\right)}{572}$$
$$+ \frac{7\sqrt{715}G_y^\alpha xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572} + \frac{21\sqrt{429}G_y^\beta xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572}$$
$$+ \frac{7\sqrt{715}G_z^\alpha xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572} - \frac{21\sqrt{429}G_z^\beta xy\left(x-y\right)\left(x+y\right)\left(x^2+y^2-10z^2\right)}{572}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B,2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{429}G_{xyz}xy\left(x-y\right)\left(x+y\right)\left(x^2+y^2-10z^2\right)}{572} + \frac{7\sqrt{715}G_x^\alpha xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$
$$+ \frac{7\sqrt{429}G_x^\beta xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144} - \frac{7\sqrt{715}G_y^\alpha yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$
$$- \frac{7\sqrt{429}G_y^\beta yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144} + \frac{7\sqrt{715}G_z^\alpha\left(x-y\right)\left(x+y\right)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$
$$- \frac{\sqrt{429}G_z^\beta\left(13x^6-45x^4y^2-150x^4z^2-45x^2y^4+540x^2y^2z^2+60x^2z^4+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2} - \frac{3\sqrt{2}ix^2y}{4} - \frac{3\sqrt{2}xy^2}{4} - \frac{3\sqrt{2}xz^2}{4} + \frac{\sqrt{2}iy^3}{2} - \frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{6}iG_x^\alpha z}{8} - \frac{\sqrt{10}iG_x^\beta z}{8} - \frac{\sqrt{6}G_y^\alpha z}{8} - \frac{\sqrt{10}G_y^\beta z}{8} - \frac{\sqrt{6}iG_z^\alpha\left(x+iy\right)}{8} - \frac{\sqrt{10}iG_z^\beta\left(x-iy\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E^{(a)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4} + \frac{\sqrt{30}xy^2}{4} - \frac{\sqrt{30}xz^2}{4} - \frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{6}G_{xyz}\left(x-iy\right)}{6} - \frac{\sqrt{10}iG_x^\alpha z}{8} + \frac{\sqrt{6}iG_x^\beta z}{24} + \frac{\sqrt{10}G_y^\alpha z}{8} + \frac{\sqrt{6}G_y^\beta z}{24} - \frac{\sqrt{10}iG_z^\alpha\left(x+iy\right)}{8} - \frac{\sqrt{6}iG_z^\beta\left(x-iy\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2} - \frac{3\sqrt{2}ix^2y}{4} - \frac{3\sqrt{2}xy^2}{4} - \frac{3\sqrt{2}xz^2}{4} + \frac{\sqrt{2}iy^3}{2} - \frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{5}iG_{xyz}\left(x^2y+ixy^2-ixz^2-yz^2\right)}{4} + \frac{\sqrt{3}iG_x^\alpha z\left(3x^2+3y^2-2z^2\right)}{12} + \frac{\sqrt{5}iG_x^\beta z\left(x-iy\right)^2}{4} - \frac{\sqrt{3}G_y^\alpha z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$- \frac{\sqrt{5}G_y^\beta z\left(x-iy\right)^2}{4} + \frac{\sqrt{3}iG_z^\alpha\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{12} - \frac{\sqrt{5}G_z^\beta\left(x^2y+ixy^2-ixz^2-yz^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E^{(a)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{3}G_{xyz}\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{12}-\frac{\sqrt{5}iG_x^{\alpha}z\left(x-iy\right)^2}{4}+\frac{\sqrt{3}iG_x^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}G_y^{\alpha}z\left(x-iy\right)^2}{4}$$
$$+\frac{\sqrt{3}G_y^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}G_z^{\alpha}\left(x^2y-ixy^2+ixz^2-yz^2\right)}{4}+\frac{\sqrt{3}iG_z^{\beta}\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{7\sqrt{110}iG_{xyz}\left(x^4y-2ix^3y^2+2ix^3z^2-2x^2y^3+ixy^4-ixz^4+2y^3z^2-yz^4\right)}{88}+\frac{5\sqrt{66}iG_x^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{528}$$
$$+\frac{\sqrt{110}iG_x^{\beta}z\left(4x^4-56ix^3y-27x^2y^2+x^2z^2+28ixy^3+28ixyz^2-10y^4+29y^2z^2-3z^4\right)}{176}$$
$$-\frac{5\sqrt{66}G_y^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{528}-\frac{\sqrt{110}G_y^{\beta}z\left(10x^4+28ix^3y+27x^2y^2-29x^2z^2-56ixy^3+28ixyz^2-4y^4-y^2z^2+3z^4\right)}{176}$$
$$+\frac{5\sqrt{66}iG_z^{\alpha}\left(x^5-6ix^4y-5x^3y^2-5x^3z^2-5ix^2y^3+51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4+iy^5-5iy^3z^2-6iyz^4\right)}{528}$$
$$-\frac{\sqrt{110}iG_z^{\beta}\left(3x^5-10ix^4y-29x^3y^2-x^3z^2+29ix^2y^3-27ix^2yz^2+10xy^4+27xy^2z^2-4xz^4-3iy^5+iy^3z^2+4iyz^4\right)}{176}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E^{(a)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{66}G_{xyz}\left(8x^5-15ix^4y-40x^3y^2-40x^3z^2+40ix^2y^3-30ix^2yz^2+15xy^4+30xy^2z^2+15xz^4-8iy^5+40iy^3z^2-15iyz^4\right)}{264}$$
$$-\frac{\sqrt{110}iG_x^{\alpha}z\left(10x^4-56ix^3y+27x^2y^2-29x^2z^2+28ixy^3+28ixyz^2-4y^4-y^2z^2+3z^4\right)}{176}$$
$$-\frac{\sqrt{66}iG_x^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{528}$$
$$-\frac{\sqrt{110}G_y^{\alpha}z\left(4x^4+28ix^3y-27x^2y^2+x^2z^2-56ixy^3+28ixyz^2-10y^4+29y^2z^2-3z^4\right)}{176}$$
$$-\frac{\sqrt{66}G_y^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{528}$$
$$-\frac{\sqrt{110}iG_z^{\alpha}\left(3x^5-4ix^4y-x^3y^2-29x^3z^2-ix^2y^3+27ix^2yz^2-4xy^4+27xy^2z^2+10xz^4+3iy^5-29iy^3z^2+10iyz^4\right)}{176}$$
$$+\frac{\sqrt{66}iG_z^{\beta}\left(11x^5-60ix^4y-55x^3y^2-55x^3z^2+55ix^2y^3+195ix^2yz^2+60xy^4-195xy^2z^2+60xz^4-11iy^5+55iy^3z^2-60iyz^4\right)}{528}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_x^{\alpha}}{2}-\frac{\sqrt{2}iG_y^{\alpha}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E^{(a)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_x^{\beta}}{2}+\frac{\sqrt{2}iG_y^{\beta}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}iG_{xyz}z\left(x+iy\right)}{2}+\frac{\sqrt{30}G_x^{\alpha}\left(8x^2+3ixy-4y^2-4z^2\right)}{120}-\frac{\sqrt{2}iG_x^{\beta}\left(xy-2iy^2+2iz^2\right)}{8}$$
$$+\frac{\sqrt{30}iG_y^{\alpha}\left(4x^2+3ixy-8y^2+4z^2\right)}{120}-\frac{\sqrt{2}iG_y^{\beta}\left(2x^2-ixy-2z^2\right)}{8}-\frac{\sqrt{30}G_z^{\alpha}z\left(x-iy\right)}{40}+\frac{\sqrt{2}G_z^{\beta}z\left(x+iy\right)}{8}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E^{(a)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{2}iG_x^{\alpha}\left(xy-2iy^2+2iz^2\right)}{8}-\frac{\sqrt{30}iG_x^{\beta}xy}{8}+\frac{\sqrt{2}iG_y^{\alpha}\left(2x^2-ixy-2z^2\right)}{8}-\frac{\sqrt{30}G_y^{\beta}xy}{8}-\frac{\sqrt{2}G_z^{\alpha}z\left(x-iy\right)}{8}-\frac{\sqrt{30}G_z^{\beta}z\left(x+iy\right)}{8}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{165}iG_{xyz}z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{132}+\frac{\sqrt{11}G_x^{\alpha}\left(8x^4+10ix^3y-24x^2y^2-24x^2z^2+10ixy^3-60ixyz^2+3y^4+6y^2z^2+3z^4\right)}{88}$$
$$+\frac{\sqrt{165}iG_x^{\beta}\left(18x^3y+6ix^2y^2-6ix^2z^2-10xy^3-24xyz^2-iy^4+iz^4\right)}{264}$$
$$-\frac{\sqrt{11}iG_y^{\alpha}\left(3x^4-10ix^3y-24x^2y^2+6x^2z^2-10ixy^3+60ixyz^2+8y^4-24y^2z^2+3z^4\right)}{88}$$
$$+\frac{\sqrt{165}iG_y^{\beta}\left(x^4+10ix^3y-6x^2y^2-18ixy^3+24ixyz^2+6y^2z^2-z^4\right)}{264}$$
$$-\frac{5\sqrt{11}G_z^{\alpha}z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{44}+\frac{\sqrt{165}G_z^{\beta}z\left(5x^3+12ix^2y+12xy^2-9xz^2+5iy^3-9iyz^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E^{(a)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{35\sqrt{11}iG_{xyz}z\left(x^3-xz^2+iy^3-iyz^2\right)}{132}-\frac{\sqrt{165}iG_x^{\alpha}\left(10x^3y-6ix^2y^2+6ix^2z^2-18xy^3+24xyz^2+iy^4-iz^4\right)}{264}$$
$$-\frac{\sqrt{11}G_x^{\beta}\left(28x^4+10ix^3y-84x^2y^2-84x^2z^2+10ixy^3-60ixyz^2-7y^4+126y^2z^2-7z^4\right)}{264}$$
$$-\frac{\sqrt{165}iG_y^{\alpha}\left(x^4-18ix^3y-6x^2y^2+10ixy^3+24ixyz^2+6y^2z^2-z^4\right)}{264}$$
$$+\frac{\sqrt{11}iG_y^{\beta}\left(7x^4+10ix^3y+84x^2y^2-126x^2z^2+10ixy^3-60ixyz^2-28y^4+84y^2z^2+7z^4\right)}{264}$$
$$+\frac{\sqrt{165}G_z^{\alpha}z\left(9x^3+12ix^2y-12xy^2-5xz^2-9iy^3+5iyz^2\right)}{132}-\frac{5\sqrt{11}G_z^{\beta}z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{7\sqrt{1430}iG_{xyz}z\left(x^5+16ix^4y-16x^3y^2+2x^3z^2-16ix^2y^3-16ix^2yz^2+16xy^4-16xy^2z^2+xz^4+iy^5+2iy^3z^2+iyz^4\right)}{1144}$$
$$+\frac{5\sqrt{858}G_x^{\alpha}\left(16x^6+42ix^5y-120x^4y^2-120x^4z^2-147ix^3y^3+21ix^3yz^2+90x^2y^4+180x^2y^2z^2+90x^2z^4+42ixy^5+21ixy^3z^2-21ixyz^4-5y^6-15y^4z^2-15\right.}{6864}$$
$$-\frac{7\sqrt{1430}iG_x^{\beta}\left(4x^5y+16ix^4y^2-16ix^4z^2-19x^3y^3+17x^3yz^2-16ix^2y^4+16ix^2z^4+10xy^5-43xy^3z^2+13xyz^4+iy^6+iy^4z^2-iy^2z^4-iz^6\right)}{2288}$$
$$+\frac{5\sqrt{858}iG_y^{\alpha}\left(5x^6+42ix^5y-90x^4y^2+15x^4z^2-147ix^3y^3+21ix^3yz^2+120x^2y^4-180x^2y^2z^2+15x^2z^4+42ixy^5+21ixy^3z^2-21ixyz^4-16y^6+120y^4z^2-9\right.}{6864}$$
$$+\frac{7\sqrt{1430}iG_y^{\beta}\left(x^6+10ix^5y-16x^4y^2+x^4z^2-19ix^3y^3-43ix^3yz^2+16x^2y^4-x^2z^4+4ixy^5+17ixy^3z^2+13ixyz^4-16y^4z^2+16y^2z^4-z^6\right)}{2288}$$
$$-\frac{35\sqrt{858}G_z^{\alpha}z\left(2x^5+ix^4y+x^3y^2-7x^3z^2-ix^2y^3-ix^2yz^2-xy^4+xy^2z^2+2xz^4-2iy^5+7iy^3z^2-2iyz^4\right)}{2288}$$
$$+\frac{7\sqrt{1430}G_z^{\beta}z\left(10x^5+13ix^4y-43x^3y^2-19x^3z^2-43ix^2y^3+17ix^2yz^2+13xy^4+17xy^2z^2+4xz^4+10iy^5-19iy^3z^2+4iyz^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E^{(a)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{21\sqrt{858}iG_{xyz}z\left(x^5-10x^3y^2-10ix^2y^3+10ix^2yz^2+10xy^2z^2-xz^4+iy^5-iyz^4\right)}{1144}$$
$$-\frac{7\sqrt{1430}iG_x^\alpha\left(10x^5y+16ix^4y^2-16ix^4z^2-19x^3y^3-43x^3yz^2-16ix^2y^4+16ix^2z^4+4xy^5+17xy^3z^2+13xyz^4+iy^6+iy^4z^2-iy^2z^4-iz^6\right)}{2288}$$
$$+\frac{\sqrt{858}G_x^\beta\left(8x^6+28ix^5y-60x^4y^2-60x^4z^2-175ix^3y^3+245ix^3yz^2+150x^2y^4-540x^2y^2z^2+150x^2z^4+28ixy^5+245ixy^3z^2-245ixyz^4-13y^6+45y^4z^2+\right.}{2288}$$
$$-\frac{7\sqrt{1430}iG_y^\alpha\left(x^6+4ix^5y-16x^4y^2+x^4z^2-19ix^3y^3+17ix^3yz^2+16x^2y^4-x^2z^4+10ixy^5-43ixy^3z^2+13ixyz^4-16y^4z^2+16y^2z^4-z^6\right)}{2288}$$
$$-\frac{\sqrt{858}iG_y^\beta\left(13x^6+28ix^5y-150x^4y^2-45x^4z^2-175ix^3y^3+245ix^3yz^2+60x^2y^4+540x^2y^2z^2-45x^2z^4+28ixy^5+245ixy^3z^2-245ixyz^4-8y^6+60y^4z^2-\right.}{2288}$$
$$-\frac{7\sqrt{1430}G_z^\alpha z\left(4x^5-13ix^4y+17x^3y^2-19x^3z^2-17ix^2y^3+43ix^2yz^2+13xy^4-43xy^2z^2+10xz^4-4iy^5+19iy^3z^2-10iyz^4\right)}{2288}$$
$$+\frac{7\sqrt{858}G_z^\beta z\left(4x^5-35ix^4y+35x^3y^2-25x^3z^2+35ix^2y^3+35ix^2yz^2-35xy^4+35xy^2z^2+4xz^4+4iy^5-25iy^3z^2+4iyz^4\right)}{2288}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{6}iG_x^\alpha z}{8}+\frac{\sqrt{10}iG_x^\beta z}{8}-\frac{\sqrt{6}G_y^\alpha z}{8}-\frac{\sqrt{10}G_y^\beta z}{8}+\frac{\sqrt{6}iG_z^\alpha(x-iy)}{8}+\frac{\sqrt{10}iG_z^\beta(x+iy)}{8}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{6}G_{xyz}(x+iy)}{6}+\frac{\sqrt{10}iG_x^\alpha z}{8}-\frac{\sqrt{6}iG_x^\beta z}{24}+\frac{\sqrt{10}G_y^\alpha z}{8}+\frac{\sqrt{6}G_y^\beta z}{24}+\frac{\sqrt{10}iG_z^\alpha(x-iy)}{8}+\frac{\sqrt{6}iG_z^\beta(x+iy)}{24}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{5}iG_{xyz}\left(x^2y-ixy^2+ixz^2-yz^2\right)}{4}-\frac{\sqrt{3}iG_x^\alpha z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}iG_x^\beta z(x+iy)^2}{4}-\frac{\sqrt{3}G_y^\alpha z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{5}G_y^\beta z(x+iy)^2}{4}-\frac{\sqrt{3}iG_z^\alpha\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{12}-\frac{\sqrt{5}G_z^\beta\left(x^2y-ixy^2+ixz^2-yz^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{3}G_{xyz}\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{12}+\frac{\sqrt{5}iG_x^\alpha z(x+iy)^2}{4}-\frac{\sqrt{3}iG_x^\beta z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}G_y^\alpha z(x+iy)^2}{4}$$
$$+\frac{\sqrt{3}G_y^\beta z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}G_z^\alpha\left(x^2y+ixy^2-ixz^2-yz^2\right)}{4}-\frac{\sqrt{3}iG_z^\beta\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{7\sqrt{110}iG_{xyz}\left(x^4y+2ix^3y^2-2ix^3z^2-2x^2y^3-ixy^4+ixz^4+2y^3z^2-yz^4\right)}{88}-\frac{5\sqrt{66}iG_x^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{528}$$
$$-\frac{\sqrt{110}iG_x^{\beta}z\left(4x^4+56ix^3y-27x^2y^2+x^2z^2-28ixy^3-28ixyz^2-10y^4+29y^2z^2-3z^4\right)}{176}$$
$$-\frac{5\sqrt{66}G_y^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{528}-\frac{\sqrt{110}G_y^{\beta}z\left(10x^4-28ix^3y+27x^2y^2-29x^2z^2+56ixy^3-28ixyz^2-4y^4-y^2z^2+3z^4\right)}{176}$$
$$-\frac{5\sqrt{66}iG_z^{\alpha}\left(x^5+6ix^4y-5x^3y^2-5x^3z^2+5ix^2y^3-51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4-iy^5+5iy^3z^2+6iyz^4\right)}{528}$$
$$+\frac{\sqrt{110}iG_z^{\beta}\left(3x^5+10ix^4y-29x^3y^2-x^3z^2-29ix^2y^3+27ix^2yz^2+10xy^4+27xy^2z^2-4xz^4+3iy^5-iy^3z^2-4iyz^4\right)}{176}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{66}G_{xyz}\left(8x^5+15ix^4y-40x^3y^2-40x^3z^2-40ix^2y^3+30ix^2yz^2+15xy^4+30xy^2z^2+15xz^4+8iy^5-40iy^3z^2+15iyz^4\right)}{264}$$
$$+\frac{\sqrt{110}iG_x^{\alpha}z\left(10x^4+56ix^3y+27x^2y^2-29x^2z^2-28ixy^3-28ixyz^2-4y^4-y^2z^2+3z^4\right)}{176}$$
$$+\frac{\sqrt{66}iG_x^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{528}$$
$$-\frac{\sqrt{110}G_y^{\alpha}z\left(4x^4-28ix^3y-27x^2y^2+x^2z^2+56ixy^3-28ixyz^2-10y^4+29y^2z^2-3z^4\right)}{176}$$
$$-\frac{\sqrt{66}G_y^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{528}$$
$$+\frac{\sqrt{110}iG_z^{\alpha}\left(3x^5+4ix^4y-x^3y^2-29x^3z^2+ix^2y^3-27ix^2yz^2-4xy^4+27xy^2z^2+10xz^4-3iy^5+29iy^3z^2-10iyz^4\right)}{176}$$
$$-\frac{\sqrt{66}iG_z^{\beta}\left(11x^5+60ix^4y-55x^3y^2-55x^3z^2-55ix^2y^3-195ix^2yz^2+60xy^4-195xy^2z^2+60xz^4+11iy^5-55iy^3z^2+60iyz^4\right)}{528}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_x^{\alpha}}{2}+\frac{\sqrt{2}iG_y^{\alpha}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E^{(b)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_x^{\beta}}{2}-\frac{\sqrt{2}iG_y^{\beta}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{2}iG_{xyz}z\left(x-iy\right)}{2}+\frac{\sqrt{30}G_x^{\alpha}\left(8x^2-3ixy-4y^2-4z^2\right)}{120}+\frac{\sqrt{2}iG_x^{\beta}\left(xy+2iy^2-2iz^2\right)}{8}$$
$$-\frac{\sqrt{30}iG_y^{\alpha}\left(4x^2-3ixy-8y^2+4z^2\right)}{120}+\frac{\sqrt{2}iG_y^{\beta}\left(2x^2+ixy-2z^2\right)}{8}-\frac{\sqrt{30}G_z^{\alpha}z\left(x+iy\right)}{40}+\frac{\sqrt{2}G_z^{\beta}z\left(x-iy\right)}{8}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E^{(b)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}iG_x^\alpha\left(xy+2iy^2-2iz^2\right)}{8}+\frac{\sqrt{30}iG_x^\beta xy}{8}-\frac{\sqrt{2}iG_y^\alpha\left(2x^2+ixy-2z^2\right)}{8}-\frac{\sqrt{30}G_y^\beta xy}{8}-\frac{\sqrt{2}G_z^\alpha z\left(x+iy\right)}{8}-\frac{\sqrt{30}G_z^\beta z\left(x-iy\right)}{8}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{165}iG_{xyz}z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{132}+\frac{\sqrt{11}G_x^\alpha\left(8x^4-10ix^3y-24x^2y^2-24x^2z^2-10ixy^3+60ixyz^2+3y^4+6y^2z^2+3z^4\right)}{88}$$
$$-\frac{\sqrt{165}iG_x^\beta\left(18x^3y-6ix^2y^2+6ix^2z^2-10xy^3-24xyz^2+iy^4-iz^4\right)}{264}$$
$$+\frac{\sqrt{11}iG_y^\alpha\left(3x^4+10ix^3y-24x^2y^2+6x^2z^2+10ixy^3-60ixyz^2+8y^4-24y^2z^2+3z^4\right)}{88}$$
$$-\frac{\sqrt{165}iG_y^\beta\left(x^4-10ix^3y-6x^2y^2+18ixy^3-24ixyz^2+6y^2z^2-z^4\right)}{264}$$
$$-\frac{5\sqrt{11}G_z^\alpha z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{44}+\frac{\sqrt{165}G_z^\beta z\left(5x^3-12ix^2y+12xy^2-9xz^2-5iy^3+9iyz^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E^{(b)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{35\sqrt{11}iG_{xyz}z\left(x^3-xz^2-iy^3+iyz^2\right)}{132}+\frac{\sqrt{165}iG_x^\alpha\left(10x^3y+6ix^2y^2-6ix^2z^2-18xy^3+24xyz^2-iy^4+iz^4\right)}{264}$$
$$-\frac{\sqrt{11}G_x^\beta\left(28x^4-10ix^3y-84x^2y^2-84x^2z^2-10ixy^3+60ixyz^2-7y^4+126y^2z^2-7z^4\right)}{264}$$
$$+\frac{\sqrt{165}iG_y^\alpha\left(x^4+18ix^3y-6x^2y^2-10ixy^3-24ixyz^2+6y^2z^2-z^4\right)}{264}$$
$$-\frac{\sqrt{11}iG_y^\beta\left(7x^4-10ix^3y+84x^2y^2-126x^2z^2-10ixy^3+60ixyz^2-28y^4+84y^2z^2+7z^4\right)}{264}$$
$$+\frac{\sqrt{165}G_z^\alpha z\left(9x^3-12ix^2y-12xy^2-5xz^2+9iy^3-5iyz^2\right)}{132}-\frac{5\sqrt{11}G_z^\beta z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{7\sqrt{1430}iG_{xyz}z\left(x^5-16ix^4y-16x^3y^2+2x^3z^2+16ix^2y^3+16ix^2yz^2+16xy^4-16xy^2z^2+xz^4-iy^5-2iy^3z^2-iyz^4\right)}{1144}$$
$$+\frac{5\sqrt{858}G_x^\alpha\left(16x^6-42ix^5y-120x^4y^2-120x^4z^2+147ix^3y^3-21ix^3yz^2+90x^2y^4+180x^2y^2z^2+90x^2z^4-42ixy^5-21ixy^3z^2+21ixyz^4-5y^6-15y^4z^2-15\right.}{6864}$$
$$+\frac{7\sqrt{1430}iG_x^\beta\left(4x^5y-16ix^4y^2+16ix^4z^2-19x^3y^3+17x^3yz^2+16ix^2y^4-16ix^2z^4+10xy^5-43xy^3z^2+13xyz^4-iy^6-iy^4z^2+iy^2z^4+iz^6\right)}{2288}$$
$$-\frac{5\sqrt{858}iG_y^\alpha\left(5x^6-42ix^5y-90x^4y^2+15x^4z^2+147ix^3y^3-21ix^3yz^2+120x^2y^4-180x^2y^2z^2+15x^2z^4-42ixy^5-21ixy^3z^2+21ixyz^4-16y^6+120y^4z^2-9\right.}{6864}$$
$$-\frac{7\sqrt{1430}iG_y^\beta\left(x^6-10ix^5y-16x^4y^2+x^4z^2+19ix^3y^3+43ix^3yz^2+16x^2y^4-x^2z^4-4ixy^5-17ixy^3z^2-13ixyz^4-16y^4z^2+16y^2z^4-z^6\right)}{2288}$$
$$-\frac{35\sqrt{858}G_z^\alpha z\left(2x^5-ix^4y+x^3y^2-7x^3z^2+ix^2y^3+ix^2yz^2-xy^4+xy^2z^2+2xz^4+2iy^5-7iy^3z^2+2iyz^4\right)}{2288}$$
$$+\frac{7\sqrt{1430}G_z^\beta z\left(10x^5-13ix^4y-43x^3y^2-19x^3z^2+43ix^2y^3-17ix^2yz^2+13xy^4+17xy^2z^2+4xz^4-10iy^5+19iy^3z^2-4iyz^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E^{(b)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$
\begin{aligned}
&-\frac{21\sqrt{858}iG_{xyz}z\left(x^5-10x^3y^2+10ix^2y^3-10ix^2yz^2+10xy^2z^2-xz^4-iy^5+iyz^4\right)}{1144}\\
&+\frac{7\sqrt{1430}iG_x^{\alpha}\left(10x^5y-16ix^4y^2+16ix^4z^2-19x^3y^3-43x^3yz^2+16ix^2y^4-16ix^2z^4+4xy^5+17xy^3z^2+13xyz^4-iy^6-iy^4z^2+iy^2z^4+iz^6\right)}{2288}\\
&+\frac{\sqrt{858}G_x^{\beta}\left(8x^6-28ix^5y-60x^4y^2-60x^4z^2+175ix^3y^3-245ix^3yz^2+150x^2y^4-540x^2y^2z^2+150x^2z^4-28ixy^5-245ixy^3z^2+245ixyz^4-13y^6+45y^4z^2+\right.}{2288}\\
&+\frac{7\sqrt{1430}iG_y^{\alpha}\left(x^6-4ix^5y-16x^4y^2+x^4z^2+19ix^3y^3-17ix^3yz^2+16x^2y^4-x^2z^4-10ixy^5+43ixy^3z^2-13ixyz^4-16y^4z^2+16y^2z^4-z^6\right)}{2288}\\
&+\frac{\sqrt{858}iG_y^{\beta}\left(13x^6-28ix^5y-150x^4y^2-45x^4z^2+175ix^3y^3-245ix^3yz^2+60x^2y^4+540x^2y^2z^2-45x^2z^4-28ixy^5-245ixy^3z^2+245ixyz^4-8y^6+60y^4z^2-\right.}{2288}\\
&-\frac{7\sqrt{1430}G_z^{\alpha}z\left(4x^5+13ix^4y+17x^3y^2-19x^3z^2+17ix^2y^3-43ix^2yz^2+13xy^4-43xy^2z^2+10xz^4+4iy^5-19iy^3z^2+10iyz^4\right)}{2288}\\
&+\frac{7\sqrt{858}G_z^{\beta}z\left(4x^5+35ix^4y+35x^3y^2-25x^3z^2-35ix^2y^3-35ix^2yz^2-35xy^4+35xy^2z^2+4xz^4-4iy^5+25iy^3z^2-4iyz^4\right)}{2288}
\end{aligned}
$$